

Retained-State Phase Calculus: Projection Loss, Balanced Refinement, and Certified Numerical Enclosures

Justin K. Lietz
Neuroca, Inc.
justin@neuroca.ai
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Abstract. This paper presents Phase Calculus as a retained-state grammar. The native object is a lifted state Ξ , and visible values are obtained only by projection. We prove that visible projection loses state, that the quarter-memory word Q^4 preserves visible projection while incrementing completed-turn memory, that balanced refinement follows the Fibonacci corridor, and that the canonical anchor (55,89) carries remainder 1/4895. We then separate exact algebraic claims from certified numerical enclosures for $\zeta(3)$, Catalan’s constant, Euler–Mascheroni’s constant, and the five Bring quintic roots. The examples share a single principle: retained coordinates carry the burden of proof until a projected value is justified.

Keywords: Phase Calculus, retained state, projection, Fibonacci refinement, numerical enclosure, Bring quintic.

1. Introduction

A scalar readout can hide the state that produced it. Phase Calculus is built around that observation. Its native object is not a visible number alone but a retained state carrying visible phase, arithmetic refinement, completed-turn memory, and a completion interval. The point is not cosmetic bookkeeping. It changes what can be proved, transported, and certified.

The paper has one narrow target. It states the retained-state mechanism cleanly, proves the exact algebraic claims that belong to it, and classifies the numerical examples by the strength of their evidence. The constants $\zeta(3)$, Catalan’s constant G , and Euler–Mascheroni’s constant γ are not treated as corridor-refined roots. They are direct real lifted states whose values are enclosed by independent numerical methods. The Bring quintic roots are treated separately: there the corridor is useful because a polynomial residual and a projected root can be checked together.

The retained-state source used here is the Phase Calculus lifted object and operator core; the canonical anchor and arithmetic-completion coefficient are inherited from the full lifted object formalism; the Bring quintic numerical example follows the unified quintic surface. The purpose of this paper is to put those ingredients into a standard theorem/proof format with the numerical material clearly marked as numerical enclosure. A

companion archive contains the Lean4 proof scripts for the exact finite statements, SymPy symbolic checks, and executable notebooks for the numerical enclosures.

2. Retained state and projection

Definition 2.1 (Lifted state). The retained Phase Calculus state is

$$\Xi = (A, q, \theta, \kappa, c), \quad q = (u, v), \quad 1 \leq u \leq v,$$

with completion interval

$$c = C(\theta, 1/(uv)) = [\theta - \pi/(uv), \theta + \pi/(uv)].$$

The visible projection is

$$\Pi_{\text{vis}}(\Xi) = A e^{i\theta}.$$

Definition 2.2 (Primitive operators). The finite operator core is $\{Q, B, L\}$. The quarter operator changes phase by $\pi/2$. The balanced refinement operator acts on the arithmetic carrier by

$$B(u, v) = \text{sort}(v, u+v).$$

The host-lift operator changes the host coordinate and translates phase by one quarter turn. Only Q and B are needed for the statements below.

Theorem 2.3 (Visible projection is not state-complete). There exist distinct retained states $\Xi_0 \neq \Xi_1$ such that $\Pi_{\text{vis}}(\Xi_0) = \Pi_{\text{vis}}(\Xi_1)$.

Proof. Choose two states with the same amplitude and arithmetic carrier but with phases θ and $\theta + 2\pi$. Then $A e^{i(\theta+2\pi)} = A e^{i\theta}$. The projected values agree. The retained states differ because the completed-turn memory changes from κ to $\kappa + 1$. Hence the visible projection is not state-complete.

Theorem 2.4 (Quarter-memory word). For every retained state,

$$\Pi_{\text{vis}}(Q^4 \Xi) = \Pi_{\text{vis}}(\Xi), \quad \kappa(Q^4 \Xi) = \kappa(\Xi) + 1.$$

More generally, $(Q^4)^m$ preserves visible projection and increments κ by m .

Proof. Each application of Q adds $\pi/2$ to the phase. Four applications add 2π , so the visible projection is unchanged. The register κ records completed 2π turns, so one block Q^4 increments it by one. Iteration gives the general statement.

3. Balanced refinement and the canonical anchor

Definition 3.1 (Fibonacci convention). Let $F_1=1$, $F_2=1$, and $F_{n+2}=F_{n+1}+F_n$.

Theorem 3.2 (Balanced Fibonacci corridor). Starting from $(u_0, v_0)=(1, 1)$ and iterating $B(u, v)=\text{sort}(v, u+v)$ gives

$$(u_n, v_n)=(F_{n+1}, F_{n+2}).$$

At depth 9 this gives

$$(u_9, v_9)=(55, 89), \quad u_9 v_9 = 4895,$$

so the carried remainder at the canonical anchor is $1/4895$.

Proof. The base case is $(u_0, v_0)=(1, 1)=(F_1, F_2)$. If $(u_n, v_n)=(F_{n+1}, F_{n+2})$, then $B(u_n, v_n)=\text{sort}(F_{n+2}, F_{n+1}+F_{n+2})=(F_{n+2}, F_{n+3})$. Induction proves the formula. The listed anchor follows by evaluating the Fibonacci sequence.

Corollary 3.3 (Depth-21 half-width). At depth 21, $(u_{21}, v_{21})=(17711, 28657)$ and

$$\delta_{21} = \pi/(17711 \cdot 28657) = 6.18979215099812 \times 10^{-9} < 10^{-8}.$$

Proof. The pair is obtained by the same Fibonacci formula. The numerical bound is direct evaluation of $\pi/(17711 \cdot 28657)$.

Proposition 3.4 (Signed two-sided coefficient). On the canonical anchor, the carried edge coefficient is $1/24$ per edge. The two-sided coefficient is therefore $1/12$, and the vacuum-signed two-sided coefficient is $-1/12$.

Proof. The anchor carries two symmetric completion edges. If each edge contributes $1/24$, summing the two sides gives $2 \cdot (1/24) = 1/12$. Applying the signed rendering gives $-1/12$.

4. Exact statements and numerical enclosures

The preceding section contains exact algebraic statements. The constants in this section are different. They are not presented as irrationality or transcendence theorems. They are numerical enclosures attached to direct real lifted states

$$\Xi_c = (|c|, (1, 1), 0, 0, [-\pi, \pi]),$$

with the sign stored by the real readout convention. The proof obligation is a numerical radius around the projected real value.

Proposition 4.1 (Apéry's constant: numerical enclosure). Let

$$S_N = (5/2) \sum_{k=1}^N (-1)^{k-1} / (k^3 C(2k, k)). \text{ For } N=92,$$

$$|\zeta(3) - S_{92}| \leq 7.23187387003819902884973268112 \times 10^{-61}.$$

Thus $\zeta(3) = 1.20205690315959428539973816151144999076498629234049888179...$ within that radius.

Proof. The accelerated Apéry series has the displayed central-binomial form. The omitted tail after N terms is bounded by $(5/2) \cdot 1/((N+1)^3 C(2N+2, N+1)) \cdot (4/3)$. Substitution of $N=92$ gives the stated radius.

Proposition 4.2 (Catalan's constant: numerical enclosure). Let

$$G_N = \sum_{k=0}^{N-1} (-1)^k / (2k+1)^2. \text{ For } N=500000,$$

$$|G - G_N| \leq 1/(2N+1)^2 = 9.99998000002999996000004999994 \times 10^{-13}.$$

The partial-sum center

$G_{500000} = 0.91596559417671901505460501493238409827414937449517...$ encloses Catalan's constant within that radius.

Proof. The displayed series is alternating with monotonically decreasing positive term magnitudes. The alternating-series estimate bounds the error by the first omitted term.

Proposition 4.3 (Euler-Mascheroni constant: numerical enclosure). Let

$$E_{n,M} = H_n - \log n - 1/(2n) + \sum_{m=1}^{M-1} B_{2m} / (2m n^{2m}).$$

With $n=1000$ and $M=10$, $|\gamma - E_{1000,10}| < 3 \times 10^{-59}$. After decimal rounding, this gives $\gamma = 0.57721566490153286060651209008240243104215933593992...$ with radius 10^{-50} .

Proof. This is the Euler-Maclaurin expansion for the harmonic numbers. The next omitted term is bounded by $|B_{20}|/(20 \cdot 1000^{20}) = 2.645621212121... \times 10^{-59}$, so the stated pre-rounding radius follows. The displayed decimal is then quoted with the larger radius 10^{-50} .

Table 1. Certified numerical enclosures for direct real lifted states. These are numerical bounds, not algebraic closed forms.

Constant	Center	Radius	Method
$\zeta(3)$	1.20205690315959428539973816151144999076	7.232×10^{-61}	Apéry tail
G	0.91596559417671901505460501493238409827	1.000×10^{-12}	Alternating tail
γ	0.57721566490153286060651209008240243104	1.000×10^{-50}	Euler-Maclaurin enclosure

5. Bring quintic numerical certification

The quintic example has a different structure. The values are not direct constants with a scalar series. They are projected roots of a polynomial, so the retained state is refined until the angular half-width and polynomial residual are both small.

Example 5.1 (Bring quintic). Consider $p(x)=x^5-x+1$. The five projected roots certified at depth 21 are listed in Table 2. Each uses the same corridor pair $(17711,28657)$ and the same half-width $6.18979215099812 \times 10^{-9}$.

Table 2. Depth-21 numerical certification of the five roots of x^5-x+1 . All rows use corridor half-width $6.18979215099812 \times 10^{-9}$; the residual column reports $|p(r)|$ for the projected root.

Index	Projected root	Residual
0	-1.1673039782614187	1.359×10^{-15}
1	$-0.1812324444698754-1.0839541013177107i$	3.953×10^{-15}
2	$-0.1812324444698754+1.0839541013177107i$	7.022×10^{-16}
3	$0.7648844336005848-0.3524715460317263i$	1.066×10^{-15}
4	$0.7648844336005848+0.3524715460317263i$	1.110×10^{-16}

These are numerical certifications: the corridor supplies a narrow angular bracket, and the polynomial residual is checked independently. No finite radical expression is claimed.

Proposition 5.2 (Depth-21 root enclosure). For the five roots displayed in Table 2, the retained corridor half-width is below 10^{-8} and every projected polynomial residual is below 4×10^{-15} .

Proof. The half-width statement follows from Corollary 3.3 and the common half-width recorded in Table 2. Direct substitution of each projected root into $p(x)=x^5-x+1$ gives the residuals in Table 2, all below 4×10^{-15} .

6. Unifying principle

The examples have different numerical mechanisms, but the retained-state principle is the same. Projection is delayed until the state has carried the information required for the requested readout. For algebraic transport, the carrier keeps completed turns,

arithmetic refinement, and branch data that visible scalar readout forgets. For direct constants, the carrier state is trivial and the useful burden is the certified numerical radius. For polynomial roots, the carrier state supplies angular refinement, while the polynomial residual supplies the independent projected check.

This is the correct separation. The corridor is not a universal replacement for every analytic series. It is a retained-state refinement law. Constants with known series are best handled by direct numerical enclosures. Branch-valued algebraic problems are best handled by retained transport followed by projected residual checks.

7. Conclusion

Retained-state Phase Calculus gives a precise distinction between visible equality and state equality. The exact results are projection non-completeness, quarter-memory preservation, Fibonacci corridor refinement, the (55,89) anchor, and the depth-21 half-width bound. The numerical results are explicitly weaker but still useful: certified enclosures for $\zeta(3)$, Catalan’s constant, and Euler-Mascheroni’s constant, and depth-21 projected certification for the five roots of $x^5 - x + 1$. The common thesis is compact: compute in the retained state, then project only at the boundary where the required burden has already been carried.

Data and code availability

The companion archive includes the LaTeX source, bibliography, Lean4 proof scripts for the exact finite claims, SymPy checks for symbolic identities and numerical residuals, and an executable notebook reproducing the constant enclosures.

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Conflict of interest

The author is affiliated with Neuroca, Inc.

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